OLLSCOIL TEICNEOLAÍOCHTA BHAILE ÁTHA CLIATH
TECHNOLOGICAL UNIVERSITY DUBLIN

TECHNOLOGICAL UNIVERSITY DUBLIN
BLANCHARDSTOWN CAMPUS

Academic Term	2021-22
Year of Study	Year 2
Semester	Semester 2 – Autumn Paper
Date of Examination	
Time of Examination	

Programme Code	**Programme Title**	**Module Code**
TU653	Higher Certificate in Science in Computing in information Technology	COMP H2014
TU757	Bachelor of Science in Computing in Information Technology	COMP H2014
TU860	Bachelor of Science (Honours) in Computing	COMP H2014
TU758	Bachelor of Science in Computing in Digital Forensics & Cyber Security	COMP H2014
TU863	Bachelor of Science (Honours) in Computing in Digital Forensics & Cyber Security	COMP H2014

Module Title	Operating Systems

Internal Examiner	**Dr. Kevin Farrell**
External Examiner	**Dr. Simon Caton**

Instructions to candidates:

1.	**To ensure that you take the correct examination, please check that the module and programme which you are following is listed in the table above.**
2.	**Answer ANY four questions.**
3.	**Where a question asks you to use diagrams, to obtain full marks, you MUST provide diagrams.**
4.	**Where a question asks you to use code, to obtain full marks, you MUST provide code.**
5.	**This paper is worth 100 marks. All questions are worth 25 marks each.**
6.	**Please make sure you write the question numbers you answer in the boxes on the front of your answer book.**
7.	**This is a two hour examination.**

DO NOT TURN OVER THIS PAGE UNTIL YOU ARE TOLD TO DO SO

Question 1

If you answer this question please write the number on the front of your answer book

a) Describe the *modular kernel* methodology of operating systems design, including any benefits of this approach.

(5 marks)

b) (i) Explain the term *Interprocess Communication (IPC).*

(4 marks)

(ii) Compare and contrast the methods of *Direct Communication* and *Indirect Communciation* used in IPC.

(8 marks)

c) Explain the concept of a Remote Procedure Call (RPC), and how it is implemented. Use a diagram to aid your answer.

(8 marks)

(25 Marks Total)

Did you answer this question? Have you written the number of the question in the box on the front of your answer book?

Question 2

If you answer this question please write the number on the front of your answer book

a) Explain the following terms with regard to Process Management:
Pre-emptive scheduling, context change, multiprogramming.

(3 marks)

b) Name and describe **five** states in the *Process Life Cycle* in an operating system, and illustrate with a *Process State Diagram* showing the transitions between the different states.

(10 marks)

c) The following table is information obtained regarding **five** different processes when a particular process scheduling algorithm is used:

(a) *Process#*	(b) *Est. Run Time*	(c) *Waiting Time*	(d) *Ratio b/a*
1	2	0	0
2	60	2	0.03
3	1	62	62
4	3	63	21
5	50	66	1.32

Describe the scheduling algorithm, which would lead to this output. You should answer the following questions regarding the algorithm, giving reasons for your answer in each case:

1. Is it Pre-emptive or Non-pre-emptive?
2. What type of queue is involved?
3. What are the uses of this algorithm?
4. What are the disadvantages of this algorithm?

(12 marks)

(25 Marks Total)

Did you answer this question? Have you written the number of the question in the box on the front of your answer book?

Question 3

If you answer this question please write the number on the front of your answer book

a) Explain the concept of *multi-threaded* processes, and how they differ from single-threaded processes. Use diagram(s) to aid your answer.

(4 marks)

b) Describe any **one** of the **four** common models of threading implementation in modern computing systems.

(6 marks)

c) Describe, under the following **five** headings, the operating systems issues associated with multi-threaded processes:

i. Semantics of **fork()** and **exec()** system calls
ii. Thread Cancellation
iii. Signal Handling
iv. Thread Pools
v. Thread Specific Data

(15 marks)

(25 Marks Total)

Did you answer this question? Have you written the number of the question in the box on the front of your answer book?

Question 4

If you answer this question please write the number on the front of your answer book

a) Give a description of the *Filesystem Hierarchy Standard (FHS)* under the following headings:

   i. What is the FHS?
   ii. What are its advantages?
   iii. How is it achieved?
   iv. Who is its target audience?

(8 marks)

b) Describe in detail the **two** independent categorisations of files in the *Filesystem Hierarchy Standard* under the following headings:

   i. What are the **two** categorisations?
   ii. Features
   iii. Rationale
   iv. Historical problem(s) overcome by the categorisations

   Provide examples for each categorisation to illustrate your answer.

(10 marks)

c) According to the *Filesystem Hierarchy Standard*

   i. what is the purpose of the *root filesystem*?
   ii. Name **four** of the subdirectories, or symbolic links to subdirectories, which are required to exist in the *root directory*? Give a one sentence description of the purpose of each of the subdirectories.

(7 marks)

(25 Marks Total)

Did you answer this question? Have you written the number of the question in the box on the front of your answer book?

Question 5

If you answer this question please write the number on the front of your answer book

a) Describe the *Bounded-Buffer Producer-Consumer Problem*, detailing why a synchronisation solution is necessary.

(4 marks)

b) The code-listings below represent a proposed solution to the *Bounded-Buffer Producer-Consumer* program. Explain the problem(s) with this code with particular reference to the **`counter++`** and **`counter--`** statements.

(6 marks)

```
/* Shared Data with Initialisation */
#define BUFFER_SIZE 10
typedef struct {
     . . .
} item;

item buffer[BUFFER_SIZE];
int in = 0;
int out = 0;

/* Producer Code */
while (true)
{
     /* produce an item in next produced */

     while (counter == BUFFER_SIZE)
           ;     /* do nothing */

     buffer[in] = next_produced;
     in = (in + 1) % BUFFER_SIZE;
     counter++;
}

/* Consumer Code */
while (true)
{
     while (counter == 0)
           ;     /* do nothing */

     next_consumed = buffer[out];
     out = (out + 1) % BUFFER_SIZE;
     counter--;

     /* consume the item in next_consumed */
}
```

This Question is continued on the next page...

c) In relation to process synchronisation, explain what the *critical-section* problem is.

(5 marks)

d) Give the **three** conditions that need to be satisfied in order to solve the *critical-section* problem.

(3 marks)

e) Describe and explain the *Test-and-Set Lock (TSL)* hardware instruction, explaining how it can be used to solve the *critical-section problem* with **two** processes. Provide pseudo-code fragments to aid your answer, and explain how the code works.

(7 marks)

(25 Marks Total)

Did you answer this question? Have you written the number of the question in the box on the front of your answer book?